

Answer questions(15%)

a. char 和 varchar 的區別

char 是固定長度的字元型別，而 varchar 是可變長度的字元型別。char 會自動填充到指定長度，而 varchar 根據實際輸入長度儲存數據。

b. MySQL 中 boolean 型別如何儲存

MySQL 中沒有實際的 boolean 型別，通常用 tinyint 來儲存 boolean

c. 數字型別的大小和範圍

不同數字型別所需的位元組數和範圍：

tinyint: 1 byte, -128 ~ 127 或 0 ~ 255

smallint: 2 bytes, -32768 ~ 32767 或 0 ~ 65535

mediumint: 3 bytes, -8388608 ~ 8388607 或 0 ~ 16777215

int: 4 bytes, -2147483648 ~ 2147483647 或 0 ~ 4294967295

TASK (85%)

A.create table

```
mysql> describe customers
-> ;
+-----+-----+-----+-----+-----+-----+
| Field                | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| customer_id          | varchar(50)   | NO   | PRI | NULL    |       |
| customer_unique_id   | varchar(50)   | YES  |     | NULL    |       |
| customer_zip_code_prefix | int           | YES  |     | 0       |       |
| customer_city         | varchar(50)   | YES  |     | NULL    |       |
| customer_state        | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql>
```

```
mysql>
mysql> describe geolocation;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| geolocation_zip_code_prefix | varchar(50) | YES | | NULL | |
| geolocation_lat | float | NO | PRI | NULL | |
| geolocation_lng | float | NO | PRI | NULL | |
| geolocation_city | varchar(50) | YES | | NULL | |
| geolocation_state | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

```
mysql> describe order_items;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| order_id | varchar(50) | NO | PRI | NULL | |
| order_item_id | int | NO | PRI | 0 | |
| product_id | varchar(50) | YES | | NULL | |
| seller_id | varchar(50) | YES | | NULL | |
| shipping_limit_date | datetime | YES | | NULL | |
| price | float | YES | | NULL | |
| freight_value | float | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

```
mysql> describe orders;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| order_id | varchar(50) | NO | PRI | NULL | |
| customer_id | varchar(50) | YES | MUL | NULL | |
| order_status | varchar(50) | YES | | NULL | |
| order_purchase_timestamp | datetime | YES | | NULL | |
| order_approved_at | datetime | YES | | NULL | |
| order_delivered_carrier_date | datetime | YES | | NULL | |
| order_delivered_customer_date | datetime | YES | | NULL | |
| order_estimated_delivery_date | datetime | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

mysql>
```

```
mysql> insert into payments (order_id, payment_sequential, payment_type,
    payment_installments, payment_value) values (1, 1, 'credit_card', 1, 1);
Query OK, 0 rows affected (0.03 sec)

mysql> describe payments;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| order_id | varchar(50) | NO | PRI | NULL | |
| payment_sequential | int | YES | | 0 | |
| payment_type | varchar(50) | YES | | NULL | |
| payment_installments | int | YES | | 0 | |
| payment_value | float | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

```

+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| product_id | varchar(50) | NO | PRI | NULL | |
| product_category_name | varchar(50) | YES | | NULL | |
| product_name_length | int | YES | | 0 | |
| product_description_length | int | YES | | 0 | |
| product_photos_qty | int | YES | | 0 | |
| product_weight_g | int | YES | | 0 | |
| product_length_cm | int | YES | | 0 | |
| product_height_cm | int | YES | | 0 | |
| product_width_cm | int | YES | | 0 | |
+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)

mysql>

```

```

mysql> ALTER TABLE sellers ADD seller_state VARCHAR(50)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> describe sellers;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| seller_id | varchar(50) | NO | PRI | NULL | |
| seller_zip_code_prefix | int | YES | | 0 | |
| seller_city | varchar(50) | YES | | NULL | |
| seller_state | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)

mysql>

```

B.Query Task :

1.

```

1 row in set (0.01 sec)

mysql> SELECT COUNT(DISTINCT customer_id) AS number_of_client FROM
+-----+
| number_of_client |
+-----+
| 99441 |
+-----+
1 row in set (0.08 sec)

mysql>

```

2.

```
mysql> SELECT COUNT(DISTINCT customer_unique_id) AS number_of_diff_id FROM customers WHERE customer_unique_id LIKE 'ff9%';
+-----+
| number_of_diff_id |
+-----+
| 35 |
+-----+
1 row in set (0.01 sec)

mysql>
```

3.

```
mysql> SELECT customer_city AS city, COUNT(*) AS cnt
-> FROM customers
-> JOIN orders ON customers.customer_id = orders.customer_id
-> WHERE customer_state = 'RJ'
-> GROUP BY customer_city
-> ORDER BY cnt DESC
-> LIMIT 3;
+-----+-----+
| city          | cnt |
+-----+-----+
| rio de janeiro | 6882 |
| niteroi        | 849 |
| nova iguacu    | 442 |
+-----+-----+
3 rows in set (0.04 sec)
```

4.

```
mysql> SELECT order_id,
-> (order_delivered_customer_date - order_delivered_carrier_date) AS delivery_days
-> FROM orders
-> WHERE order_delivered_customer_date IS NOT NULL
-> AND order_delivered_carrier_date IS NOT NULL
-> ORDER BY delivery_days DESC
-> LIMIT 3;
+-----+-----+
| order_id          | delivery_days |
+-----+-----+
| 2aa91108853cecb43c84a5dc5b277475 | 20171120194447 |
| 6e6527028de694ccade37f5a15a6d84a | 9381829499 |
| ed8e9faf1b75f43ee027103957135663 | 9316987372 |
+-----+-----+
3 rows in set (0.02 sec)

mysql>
```

5.

```
mysql> SELECT customer_state AS state, COUNT(*) AS cnt
-> FROM orders
-> JOIN customers ON orders.customer_id = customers.customer_id
-> WHERE order_status != 'delivered'
-> GROUP BY customer_state;
```

state	cnt
SP	1245
RJ	502
BA	124
RS	121
MG	281
MS	14
ES	38
RN	11
SC	91
PR	122
PE	59
GO	63
CE	57
RR	5
MA	30
AL	16
AC	1
PA	29
DF	60
PB	19
SE	15
MT	21
PI	19

RO	10
TO	6
AM	3
AP	1

27 rows in set (0.22 sec)

```
mysql> █
```

6.

```
mysql> SELECT order_id
-> FROM orders
-> WHERE order_id NOT IN (SELECT order_id FROM payments);
+-----+
| order_id |
+-----+
| bfbdb0f9bdef84302105ad712db648a6c |
+-----+
1 row in set (0.07 sec)

mysql> █
```

7.

```
mysql> WITH DeliveryDays AS (
-> SELECT
->     o.order_id,
->     c.customer_state,
->     (o.order_delivered_customer_date - o.order_delivered_carrier_date) AS delivery_days,
->     ROW_NUMBER() OVER (PARTITION BY c.customer_state ORDER BY (o.order_delivered_customer_date - o.order_delivered_carrier_date) DESC) AS rn
-> FROM orders o
-> JOIN customers c ON o.customer_id = c.customer_id
-> WHERE
->     o.order_delivered_customer_date IS NOT NULL
->     AND o.order_delivered_carrier_date IS NOT NULL
-> )
-> SELECT
->     customer_state AS state,
->     order_id,
->     delivery_days
-> FROM DeliveryDays
-> WHERE rn = 1
-> ORDER BY customer_state DESC;
+-----+-----+-----+
| state | order_id | delivery_days |
+-----+-----+-----+
| TO | a523c396a29386423aa594fe448ca713 | 8900030943 |
| SP | 2aa91108853cecb43c84a5dc5b277475 | 20171120194447 |
| SE | 8a978fd62dcc5d4c42fe95a394f718bb | 8992790201 |
| SC | 1d3b8d78074e0b50f10346941e67e9de | 8980115123 |
| RS | 7c7b51735653e5eabc93b37b0f454428 | 9085000308 |
| RR | 3a9e44b449e2ec9986a329724bc824eb | 8987982630 |
| RO | 7a7afad8caa7dffd8773430c8fab7ea | 8894951919 |
| RN | 5c092582307d1169e1a475648c90003f | 8987965894 |
+-----+-----+-----+

| RJ | 6e6527028de694ccade37f5a15a6d84a | 9381829499 |
| PR | d1f789756b59a9fb40b3becadeac90e4 | 9100859445 |
| PI | 43716fe010471178ec7f9c3be2a0c505 | 8983963167 |
| PE | f967b3b4e10db096bfc62535abb40d43 | 9093937670 |
| PB | b417d51297514f6e670d1ffb3513ef53 | 8985872200 |
| PA | 5005f0fff3706e3297c1015ed5e41d18 | 8991011000 |
| MT | 7ee39ad7cefae28934df87c631928f88 | 9000969220 |
| MS | 37a63972d6fa35618d22858e3cffe1c1 | 8913052397 |
| MG | 5012f0f1b371c6d1dffb743758d3c522 | 8980964764 |
| MA | 8ff6ab5092f178654b69b95482c18556 | 8999109811 |
| GO | 43744ad15fef908c8dfcbd7fec0d786e | 8992031956 |
| ES | 4ef09dfdc36677a4d599a0c1acace4ff | 8978987310 |
| DF | 27167c20a18a136683606cb93c0168a9 | 8995980721 |
| CE | 17cc6728043d53cc948551dfbf0a338b | 9288050702 |
| BA | c5f43352377291361dad9a7366ae1c89 | 9279029513 |
| AP | 00c47fefb9a49baec12ab3654d7f548e | 8909987649 |
| AM | 1bf68d6f3bd856a3f205fdd8767fad08 | 8978961804 |
| AL | c61e3efc183db6b89f5556efb5aa8797 | 8908823804 |
| AC | feeb572a755207d889d166ca90221c84 | 8899970248 |
+-----+-----+-----+
27 rows in set (0.27 sec)
```

8.

```
mysql> SELECT COUNT(*) AS cnt
-> FROM orders
-> JOIN payments ON orders.order_id = payments.order_id
-> JOIN customers ON orders.customer_id = customers.customer_id
-> WHERE payment_type = 'debit_card'
-> AND customer_state = 'RJ'
-> AND YEAR(order_purchase_timestamp) = 2017;
+-----+
| cnt |
+-----+
| 57 |
+-----+
1 row in set (0.02 sec)

mysql> 
```

9.

```
mysql> WITH SellerSales AS (
-> SELECT seller_id, COUNT(*) AS cnt
-> FROM order_items
-> WHERE price > 150
-> GROUP BY seller_id
-> HAVING cnt > 100
-> )
-> SELECT seller_id, cnt
-> FROM SellerSales
-> ORDER BY cnt DESC
-> LIMIT 3;
+-----+-----+
| seller_id | cnt |
+-----+-----+
| 4869f7a5dfa277a7dca6462dcf3b52b2 | 780 |
| 7c67e1448b00f6e969d365cea6b010ab | 487 |
| fa1c13f2614d7b5c4749cbc52fecda94 | 476 |
+-----+-----+
3 rows in set (0.03 sec)

mysql> 
```

10.

```
mysql> WITH PaymentCounts AS (  
->     SELECT payment_type, COUNT(*) AS cnt  
->     FROM payments  
->     GROUP BY payment_type  
->     HAVING cnt > 100  
-> )  
-> SELECT payment_type, cnt  
-> FROM PaymentCounts  
-> ORDER BY cnt DESC  
-> LIMIT 2, 100;
```

```
+-----+-----+  
| payment_type | cnt |  
+-----+-----+  
| voucher      | 2739 |  
| debit_card   | 1527 |  
+-----+-----+  
2 rows in set (0.03 sec)
```

```
mysql>
```